

# Limb Motion Recognition and Assistant

## Team Structure Proposal

Distributed Software Development — UTAD × Jilin University

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### 1. Introduction

This document presents a proposed team structure for the Limb Motion Recognition and Assistant project, developed collaboratively between the Universidade de Trás-os-Montes e Alto Douro (UTAD) and Jilin University (JLU).

The proposal divides the six Project Managers and their respective teams across three technical areas (Sensor, Server, and Monitor), following a balanced 2-2-2 distribution. Each team comprises five members with clearly defined roles designed to address the challenges of cross-continental, distributed development.

This is an initial organisational proposal. The structure is intended to serve as a working baseline and may be refined as the project evolves, team dynamics are established, and technical requirements become better understood.

## 2. Rationale for the Organisational Structure

The system is divided into three functionally distinct areas Sensor, Server, and Monitor based on the following principles:

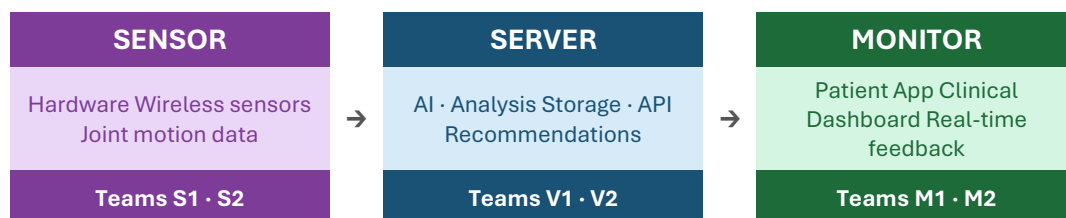
**Modularity:** Each area has a well-defined technical boundary and can be developed, tested, and iterated upon independently, provided the interface contracts between areas are agreed upon early.

**Specialisation:** The technical skills required differ significantly between areas. Sensor work demands hardware and embedded systems expertise; Server work requires backend, database, and AI knowledge; Monitor work focuses on frontend development and UX design.

**Scalability:** A modular division allows teams to scale their sub-systems without impacting the others. It also simplifies integration, as each team exposes a documented interface rather than sharing a monolithic codebase.

**Parallel development:** With a mock sensor data module produced by Team S2 in Sprint 1, all six teams can develop in parallel without waiting for physical hardware to be available. This is essential in a distributed project with a fixed timeline.

The diagram below illustrates the three areas and their relationship:



## 3. Overview of the Three Areas

### 3.1 Sensor

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The Sensor area is responsible for the physical layer of the system. Teams in this area work with wireless sensors that capture joint motion data from the patient's limb during rehabilitation exercises. Their primary output is a clean, structured data stream that feeds into the Server layer.

- **Focus:** hardware integration, firmware, wireless communication protocols.
- **Key constraint:** data format must be agreed and frozen early (Sprint 1) to unblock all Server teams.
- **Teams:** S1 (Firmware & Protocol), S2 (Data Acquisition & Mock Generator).

### 3.2 Server

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The Server area is the processing core of the system. It receives raw sensor data, applies AI-based motion recognition, stores session history, and exposes a REST/WebSocket API to the Monitor layer. It also hosts the exercise recommendation engine that analyses historical data.

- **Focus:** AI inference, data storage, API design, security, recommendation logic.
- **Key constraint:** Team V2 acts as Integration Lead from Sprint 3, coordinating end-to-end system tests.
- **Teams:** V1 (AI & Motion Recognition), V2 (API & Storage).

### 3.3 Monitor

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The Monitor area is the user-facing layer of the system. It comprises two distinct interfaces for two distinct user profiles: the patient receiving real-time rehabilitation feedback via a mobile application, and the medical professional reviewing long-term recovery data via a clinical web dashboard.

- **Focus:** real-time visualisation, 3D motion reconstruction, historical data presentation, UX.
- **Key constraint:** both Monitor teams depend on the IF2 API contract, which must be defined by V2 and M1 jointly by the end of Sprint 2.
- **Teams:** M1 (Patient Mobile App), M2 (Clinical Web Dashboard).

## 4. Team Structure

### Area: Sensor · S1 – Firmware & Wireless Protocol

#### Objective

Responsible for the physical implementation of the wireless sensors and the definition of the communication protocol used to transmit joint motion data.

#### Main Responsibilities

- Implement and calibrate the sensor firmware for joint motion capture.
- Define and implement the wireless communication protocol.
- Specify the data packet format, sampling rate, and error codes.
- Deliver the Interface Contract IF1 specification by the end of Sprint 1.

#### Team Composition

| Role                | Nationality                | Responsibility                                                                                            |
|---------------------|----------------------------|-----------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination, sprint planning, reporting to the Steering Committee.                               |
| Vice PM (Shadow PM) | Opposite nationality to PM | Leads the team on their side during the PM's off-hours; publishes daily async updates to the GitHub Wiki. |
| System Architect    | Any                        | Defines the wireless protocol, message structure, and S1–S2 internal interface.                           |
| Programmer (PT)     | Portuguese                 | Sensor firmware implementation; joint angle calibration; code review of CN programmer's work.             |
| Programmer (CN)     | Chinese                    | Hardware drivers; microcontroller integration; tests and reviews PT programmer's code.                    |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## Area: Sensor · S2 – Data Acquisition & Mock Generator

### Objective

Responsible for the data acquisition pipeline and for producing a mock sensor data stream that allows Server and Monitor teams to develop and test in parallel without depending on physical hardware.

### Main Responsibilities

- Build the data acquisition and transmission pipeline from sensor to server.
- Produce the mock sensor data module (mandatory Sprint 1 deliverable).
- Validate and sanitise incoming sensor data before forwarding.
- Co-own Interface Contract IF1 together with Team V1.

### Team Composition

| Role                | Nationality                | Responsibility                                                                                                            |
|---------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination, sprint planning, reporting to the Steering Committee.                                               |
| Vice PM (Shadow PM) | Opposite nationality to PM | Leads the team on their side during the PM's off-hours; publishes daily async updates to the GitHub Wiki.                 |
| System Architect    | Any                        | Defines IF1 (Sensor → Server contract): JSON schema, sampling rate, error handling; co-authored with V1 System Architect. |
| Programmer (PT)     | Portuguese                 | Acquisition pipeline; data validation and cleansing; code review of CN programmer's work.                                 |
| Programmer (CN)     | Chinese                    | Mock sensor data generator module; transmission layer; tests and reviews PT programmer's code.                            |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## Area: Server · V1 – AI & Motion Recognition Engine

### Objective

Responsible for integrating the Deep Learning models developed by the Jilin University research team and building the motion recognition and comparison pipeline.

### Main Responsibilities

- Integrate the external Deep Learning models for limb motion recognition.
- Build the inference pipeline that processes incoming sensor data.
- Compare recognised joint angles against expected recovery values.
- Define the internal Server API contract between V1 and V2.

### Team Composition

| Role                | Nationality                | Responsibility                                                                                                     |
|---------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination; manages integration with the external JLU Deep Learning team.                                |
| Vice PM (Shadow PM) | Opposite nationality to PM | Strategic liaison with the Jilin DL research team; leads the local side during PM's off-hours; daily async update. |
| System Architect    | Any                        | Defines the V1–V2 internal API; data models for AI results; co-owns IF1 with S2.                                   |
| Programmer (PT)     | Portuguese                 | Deep Learning model integration; inference pipeline; angle comparison logic; reviews CN programmer's code.         |
| Programmer (CN)     | Chinese                    | Sensor data pre-processing for model input; feature extraction; tests and reviews PT programmer's code.            |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## Area: Server · V2 – Backend API & Storage

### Objective

The central hub of the system. Responsible for the API gateway, data persistence, user authentication, and the exercise recommendation engine. Also acts as Integration Lead from Sprint 3.

### Main Responsibilities

- Design and implement the REST/WebSocket API gateway.
- Manage the database schema, session history, and data persistence.
- Implement the exercise recommendation logic based on historical data.
- Own Interface Contract IF2 (Server → Monitor).
- Act as Integration Lead: coordinate end-to-end integration tests from Sprint 3 onwards.

### Team Composition

| Role                | Nationality                | Responsibility                                                                                                 |
|---------------------|----------------------------|----------------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination; Integration Lead role from Sprint 3 (end-to-end system testing).                         |
| Vice PM (Shadow PM) | Opposite nationality to PM | Leads the local side during PM's off-hours; manages IF2 communications with Monitor teams; daily async update. |
| System Architect    | Any                        | Owns IF2 specification; database architecture; security and authentication model; API contract.                |
| Programmer (PT)     | Portuguese                 | API gateway; authentication; REST endpoints; recommendation engine; reviews CN programmer's code.              |
| Programmer (CN)     | Chinese                    | Database schema, queries and migrations; session data management; tests and reviews PT programmer's code.      |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## Area: Monitor · M1 – Patient Mobile Application

### Objective

Responsible for the real-time patient-facing mobile application. Patients use this app during rehabilitation exercises to receive immediate feedback on whether their movements are correct.

### Main Responsibilities

- Build the real-time mobile interface for the patient.
- Visualise reconstructed joint motion and compare with expected movement.
- Display recovery alerts and posture correction guidance.
- Integrate with V2 via WebSocket for live data streaming.

### Team Composition

| Role                | Nationality                | Responsibility                                                                                       |
|---------------------|----------------------------|------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination; defines required API endpoints with V2.                                        |
| Vice PM (Shadow PM) | Opposite nationality to PM | Leads the local side during PM's off-hours; liaison with V2 for API integration; daily async update. |
| System Architect    | Any                        | Co-owns IF2 (Monitor side); mobile app architecture; real-time UX flow design.                       |
| Programmer (PT)     | Portuguese                 | Real-time UI; motion visualisation; posture alerts; reviews CN programmer's code.                    |
| Programmer (CN)     | Chinese                    | WebSocket integration for live data; push notifications; tests and reviews PT programmer's code.     |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## Area: Monitor · M2 – Clinical Web Dashboard

### Objective

Responsible for the web-based clinical interface used by doctors and physiotherapists to review a patient's long-term recovery progress, analyse historical data, and consult medical archives.

### Main Responsibilities

- Build the clinical web dashboard for medical professionals.
- Display long-term recovery progression through charts and historical data.
- Implement 3D limb reconstruction visualisation for clinical review.
- Integrate with V2 REST API for historical data retrieval.

### Team Composition

| Role                | Nationality                | Responsibility                                                                                              |
|---------------------|----------------------------|-------------------------------------------------------------------------------------------------------------|
| Project Manager     | Portuguese or Chinese      | Overall coordination; alignment with clinical requirements.                                                 |
| Vice PM (Shadow PM) | Opposite nationality to PM | Leads the local side during PM's off-hours; daily async update.                                             |
| System Architect    | Any                        | Web dashboard architecture; data visualisation structure; clinical report templates.                        |
| Programmer (PT)     | Portuguese                 | Recovery progression dashboard; historical charts; clinical comparison views; reviews CN programmer's code. |
| Programmer (CN)     | Chinese                    | 3D limb reconstruction visualisation; V1 AI data integration; tests and reviews PT programmer's code.       |

**Cross-testing model:** Programmer (PT) formally reviews and tests the Chinese programmer's code, and vice versa. This ensures independent verification without requiring a dedicated Tester role.

## 5. Internal Team Composition with Models for Discussion

The five-member team structure defined in this proposal is common across all six teams. However, the specific distribution of roles within those five seats is open to discussion. Two possible models are presented below, each with different trade-offs regarding workload balance, technical depth, and coordination capacity.

Both models share the same foundational principles established in Section 4: the Vice PM acts as a Shadow PM with full authority on their side, and Programmers from different nationalities formally review each other's code as the primary quality-control mechanism.

### 5.1 Solution A

| Role                       | Nationality                | Responsibility / Notes                                                                                                                          |
|----------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| PM / Programmer            | Portuguese or Chinese      | Leads the team and contributes to development. Code tasks must be scoped to low-criticality modules to avoid conflict with coordination duties. |
| Vice PM / System Architect | Opposite nationality to PM | Shadow PM on the remote side. Owns architecture and interface contracts — a natural pairing since both roles require a full-system view.        |
| Programmer / Tester        | Portuguese or Chinese      | Core development. Formally cross-tests the code of the Programmer with the opposite nationality.                                                |
| Programmer / Tester        | Opposite to the row above  | Core development. Formally cross-tests their counterpart's code.                                                                                |
| Programmer / Tester        | Any (balances team ratio)  | Core development. Formally cross-tests their nearest counterpart's code.                                                                        |

### Reasoning

- **Addresses PM underutilisation:** in intermediate sprints, when coordination load is lower, the PM has real technical output rather than waiting for the next management task.
- **Strongest feature is the Vice PM / System Architect pairing:** both roles require a global system view, making the combination highly synergistic.
- **Three Programmer / Tester roles:** provide solid development capacity and distributed cross-testing.

## 5.2 Solution B

| Role                       | Nationality                | Responsibility / Notes                                                                                                                                            |
|----------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PM / System Architect      | Portuguese or Chinese      | Leads the team and owns the architecture and interface contracts. Gives the PM deep technical authority, especially useful for inter-team interface negotiations. |
| Vice PM / System Architect | Opposite nationality to PM | Shadow PM and architectural co-owner. Both PM and Vice PM hold architectural knowledge, improving resilience during timezone gaps.                                |
| Programmer / Tester        | Portuguese or Chinese      | Core development. Formally cross-tests the code of the Programmer with the opposite nationality.                                                                  |
| Programmer / Tester        | Opposite to the row above  | Core development. Formally cross-tests their counterpart's code.                                                                                                  |
| Programmer / Tester        | Any (balances team ratio)  | Core development. Formally cross-tests their nearest counterpart's code.                                                                                          |

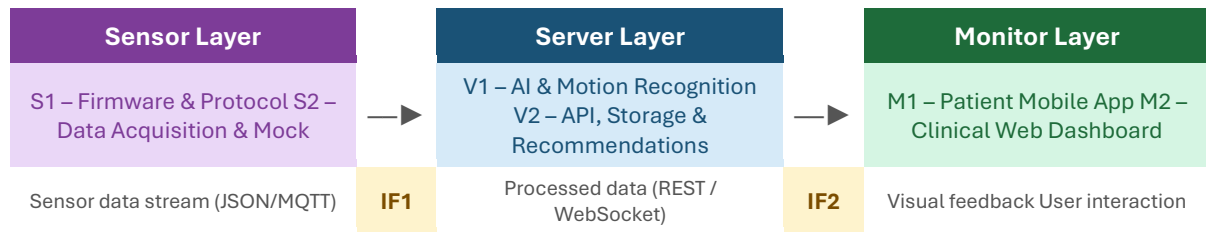
### Reasoning

- **No priority conflict for the PM:** architecture and leadership share the same type of high-level thinking. When coordination peaks, the PM is not abandoning a coding task — they are doing the same kind of work.
- **Redundant architectural knowledge:** both PM and Vice PM understand the full system design. When the PM is unreachable, the Vice PM can make all architectural decisions independently.
- **Slightly lower development throughput:** three dedicated Programmers versus the approximately 3.4 effective developers in Solution A.
- **Best suited for teams with heavy inter-team coordination:** particularly Team V2, where the PM acts as Integration Lead from Sprint 3 and must negotiate interface contracts with four other teams simultaneously.

## 6. Interaction Between Teams

### 6.1 Data Flow

The system follows a linear data flow from the physical layer to the user interface, crossing two formally defined interface boundaries:



- **IF1 - Sensor to Server:** Defines the data schema (JSON), sampling rate, error codes, and MQTT endpoint. Co-owned by the System Architects of S2 and V1. Must be frozen by the end of Sprint 1.
- **IF2 - Server to Monitor:** Defines the REST and WebSocket API specification, endpoint list, authentication tokens, and payload examples. Co-owned by the System Architects of V2 and M1. Must be frozen by the end of Sprint 2.

### 6.2 Dependencies

| From | To      | Dependency                                                              |
|------|---------|-------------------------------------------------------------------------|
| S1   | S2      | Wireless protocol specification and data packet format.                 |
| S2   | V1 + V2 | Sensor data stream via IF1. Also delivers mock data module in Sprint 1. |
| V1   | V2      | AI inference results (motion recognition, angle comparison).            |
| V2   | M1 + M2 | Processed data and analysis via IF2 REST/WebSocket API.                 |
| M1   | V2      | User session feedback and exercise interaction data.                    |

### 6.3 Coordination Mechanisms

- **Daily async standup:** Each Vice PM posts a concise update to the team's GitHub Wiki page at the end of their local working day (format: done / in progress / blockers). This replaces the need for synchronous daily meetings across timezones.
- **All-Site Meetings (ASM):** Three scheduled full-project meetings via Tencent WeMeet (VooV). Attendance is mandatory for all PMs and Vice PMs. Meeting times should rotate to share the inconvenient hour fairly.

- **Integration testing (from Sprint 3):** Led by the PM of Team V2. At minimum one cross-area integration test is required at the midpoint of the project to detect mismatches before the final sprint.
- **Git conventions:** All commit messages, pull request descriptions, and issue titles must be written in English.